ISO 9001:2008 Certified Institute

JAVA INSTITUTE FOR ADVANCED TECHNOLOGY

Department of Examinations

Unit Name	Business Component Development II
Unit Id	JIAT/BCD II
Assignment Id	JIAT/BCD II/EX/01
Assignment Summary	Students will design, implement, and evaluate an Enterprise Java Bean (EJB) application using advanced components including timer services, interceptors, transaction management, security frameworks, and professional deployment architectures. The focus is on critical analysis of EJB lifecycle management, performance optimization through exception handling, and meeting enterprise-level functional and non-functional requirements. The assignment includes comprehensive testing of EJB features, performance validation of optimization strategies, and documentation of design decisions and system improvements.
Duration	3 Weeks
Submission Via	Online (Student Portal)
Document Format	Document Format (Pdf)

GUIDELINES FOR CANDIDATES

- Your studies will be governed by the Java Institute Academic Regulations on Assessment, Progression and Awards.
- Students are expected to use reference books, the Internet, journals and other similar sources in order to accomplish the task specified above.
- Students are expected to refrain from repeating any content in their research document.
- At the re-assessment attempt, the mark is capped and the maximum mark that can be achieved is 40%.

CHEATING AND PLAGIARISM

Both cheating and plagiarism are totally unacceptable, and the Institute maintains a strict policy against them. It is YOUR responsibility to be aware of this policy and to act accordingly.

The basic principles are:

- Don’t pass off anyone else’s work as your own, including coding examples. This is plagiarism and is

viewed extremely seriously by the Institute.

- Don't submit a piece of work, in whole or in part, that has already been submitted for assessment elsewhere. This is called duplication and, like plagiarism, is viewed extremely seriously by the Institute.
- Always acknowledge all the sources you have used in your assignment or project.
- If you are using the exact words of another person, always put them in quotation marks.
- Check whether the assignment is to be completed individually or if you are allowed to work with others.
- If you are doing group work, be clear about which parts you are expected to complete on your own.
- Never make up or falsify data to support your arguments.
- Never allow others to copy your work.
- Never lend disks, memory sticks, or copies of your coursework to any other student at the Institute, as this may lead to accusations of collusion

LEARNING OUTCOMES:

This assignment addresses the following unit learning outcomes:

1. Demonstrate a clear understanding regarding the role of time services in Business Component Development while being able to create, integrate and handle the timer services of the application.

2. Identify and define the main concepts relating to Interceptor classes and methods, including the association of multiple business Interceptor methods with an enterprise bean.

3. Clearly describe transaction demarcation management, including the types of transaction, evaluating the most applicable transaction for a given scenario.

4. Undertake a critical analysis of the security architecture required in an application, in addition to the security measures taken to ensure the privacy of a session, while paying special attention to programmatic authorization.

5. Define best practices and state the benefits of using EJB technology best practices while integrating exception handling to the application for further optimization.

6. Distinguish the exception handling methods that are specific to Enterprise Java Beans and be able to optimize the performance of the application through exception handling.

7. Integrate the organisation EJB components in a split directory structure, including the package and deploy where an appropriate deployment tool as well as appropriate packaging have to be selected.

Enterprise Java Development Assignment

Global Supply Chain Management System Modernization Project

Assignment Overview:

You are a senior software architect hired by "GlobalTrade Logistics Corporation" to design a comprehensive modernization strategy for their supply chain management platform. The company needs to transition from their legacy system to a robust Java EE-based architecture that can handle enterprise-scale logistics operations while implementing automated supply chain monitoring services, comprehensive vendor data security, logistics transaction management, and optimized supply chain exception handling.

Business Scenario

GlobalTrade Logistics Corporation manages supply chains for multinational corporations across 50+ countries facing critical operational challenges:

Current State:

- Legacy system with manual shipment tracking and vendor performance monitoring
- Inconsistent logistics transaction handling causing inventory discrepancies and delivery delays
- Security vulnerabilities in vendor data access and supply chain visibility systems
- Poor exception handling leading to system failures during peak shipping seasons
- Monolithic deployment causing difficulties in updating individual supply chain modules
- Limited audit trail for customs compliance and trade regulation requirements
- Inadequate real-time tracking of shipments, inventory levels, and supplier performance

Business Requirements:

- Automated scheduling services for shipment tracking, inventory replenishment, vendor performance evaluation, customs documentation, and route optimization
- Comprehensive transaction management ensuring ACID properties across all supply chain data operations
- Multi-layered security architecture with role-based access control for logistics personnel, vendors, customs officials, and customer portals
- Robust exception handling with detailed logging and recovery mechanisms for critical supply chain disruptions
- Modular deployment architecture supporting independent updates of different supply chain service modules
- Real-time supply chain monitoring with automated alerts for shipment delays, inventory shortages, and vendor performance issues
- International trade compliance with customs regulations, import/export documentation, and trade agreement requirements
- Integration with shipping carriers, customs systems, warehouse management systems, and supplier portals

Technical Constraints:

- Must use Java/Jakarta EE platform with EJB components
- Integration with existing Supply Chain Management (SCM) database
- High availability requirement (99.9% uptime for global operations)
- Strict security compliance (international trade regulations, vendor data protection)

Your Mission

Design a complete Java EE-based supply chain architecture that addresses all business requirements while demonstrating mastery of enterprise Java concepts. Your solution must include critical analysis of timer services for logistics operations, interceptor implementation for supply chain auditing, transaction management strategies for logistics data, security architecture for global trade systems, and comprehensive exception handling with emphasis on EJB best practices and optimized deployment strategies.

Comprehensive Design Requirements

Your architecture must address enterprise supply chain functionality through integrated Java EE solutions demonstrating critical thinking and analytical evaluation of:

- **Supply Chain Timer Services Integration and Management:** Implement EJB Timer Service for automated logistics operations including shipment status updates, inventory level monitoring, vendor performance assessments, customs deadline tracking, and route optimization calculations. Critically analyse programmatic vs. declarative timer creation for supply chain workflows, evaluate timer persistence and reliability in globally distributed logistics environments, assess timeout callback method optimization for time-critical shipment alerts, and analyse timer service performance implications for supply chain efficiency and customer satisfaction.

- **Logistics Interceptor Architecture and Implementation:** Design comprehensive interceptor framework for supply chain cross-cutting concerns including logistics audit logging, vendor data validation, performance monitoring, and regulatory compliance checking (customs regulations, trade agreements, security protocols). Critically evaluate interceptor lifecycle management in global logistics contexts, analyse interceptor chain optimization for supply chain workflows, assess method-level vs. class-level interceptor strategies for different logistics processes, and evaluate interceptor performance impact on supply chain business method execution with delivery time considerations.

- **Logistics Transaction Demarcation and Management:** Implement comprehensive transaction management strategy for supply chain data operations using container-managed and bean-managed transactions. Critically analyse transaction attribute selection for different logistics scenarios (order processing, inventory updates, shipment tracking, customs documentation), evaluate transaction isolation levels for concurrent global operations, assess distributed transaction handling across multiple supply chain systems and analyse transaction performance optimization strategies with logistics data rollback scenarios.

- **Global Trade Security Architecture and Authorization:** Design multi-layered security framework incorporating declarative and programmatic security measures for supply chain data and international trade information. Critically evaluate role-based access control implementation for logistics personnel (coordinators, customs agents, warehouse managers, vendor representatives), analyse session security and management strategies for global supply chain workflows, assess authentication mechanisms for different user types and emergency logistics scenarios, and evaluate security interceptor integration with performance considerations for time-sensitive trade operations.

- **EJB Best Practices and Supply Chain Optimization:** Implement EJB components following enterprise best practices with focus on supply chain performance optimization. Critically analyse EJB component lifecycle management in logistics contexts, evaluate resource pooling strategies for supply chain applications, assess remote vs. local interface selection for carrier and vendor system integration, and analyse deployment descriptor optimization with packaging strategies for modular supply chain systems.

- **Supply Chain Exception Handling and System Resilience:** Design comprehensive exception handling framework specific to EJB environments in global logistics settings. Critically evaluate application exceptions vs. system exceptions in supply chain contexts, analyse exception propagation in distributed logistics environments, assess recovery strategies for different supply chain failure scenarios (carrier system outages, customs system failures, inventory data corruption, weather-related disruptions), and evaluate exception handling performance impact with optimization techniques ensuring supply chain continuity.
- **Supply Chain Component Organization and Deployment Strategy:** Implement split directory structure for EJB components with optimized packaging and deployment for different logistics service modules. Critically analyse JAR/WAR/EAR packaging strategies for supply chain applications, evaluate deployment descriptor management for logistics workflows, assess class loader implications for carrier and customs system integration, and analyse deployment tool selection with performance considerations for 24/7 global operations.

Non-Functional Requirements Analysis

Your solution must demonstrate critical thinking in addressing key NFRs:

- **Supply Chain Security Analysis:** Evaluate vendor data authentication and authorization strategies, analyse logistics session management approaches, assess supply chain data encryption requirements, and propose global trade security monitoring mechanisms.
- **Logistics Performance Analysis:** Evaluate timer service optimization strategies for supply chain operations, analyse interceptor performance impact on logistics workflows, assess transaction timeout optimization for trade processes, and propose performance monitoring mechanisms for global supply chain operations.
- **Supply Chain Reliability Assessment:** Evaluate exception handling strategies for logistics emergencies, analyse timer service reliability for critical supply chain processes, assess transaction recovery approaches for logistics data, and propose comprehensive availability measures for global trade systems.
- **Trade Compliance Evaluation:** Analyse supply chain audit trail requirements (customs regulations, trade agreements), evaluate international trade compliance measures, assess logistics data retention strategies, and propose compliance monitoring mechanisms for global operations.

Project Development Requirements

Develop a working supply chain prototype demonstrating practical application of Java EE technologies with measurable performance and security characteristics.

Core Implementation Requirements:

- Timer services for automated supply chain operations with persistence and clustering support for global logistics
- Interceptor framework for logistics audit, vendor validation, and supply chain performance monitoring with optimization
- Comprehensive transaction management with different attribute scenarios for supply chain data and optimization
- Multi-layered security implementation with role-based access for logistics personnel and vendor portals

- EJB components following best practices with lifecycle optimization and resource management for supply chain applications
- Advanced exception handling with recovery mechanisms and performance monitoring for logistics continuity
- Split directory structure with optimized packaging and deployment configuration for modular supply chain systems
- Database integration with connection optimization and transaction coordination for global logistics data

Technology Stack:

- Java EE 8+ (Jakarta EE acceptable)
- Application Server: WildFly, GlassFish/Payara, or WebLogic
- Database: PostgreSQL/MySQL with medical data transaction coordination
- Build Tool: Maven with multi-module structure for clinical applications
- Testing: JUnit 5 + Arquillian + supply chain security testing tools
- Security: JAAS integration with custom login modules for supply chain authentication

Deliverables

1. **Working Project Implementation**

   Source Code Package:

   - Complete supply chain application with timer services, interceptors, and security implementation
   - Logistics database schema with transaction optimization and audit trail support
   - Configuration files with supply chain security, transaction, and performance settings
   - Deployment guide with packaging and logistics security configuration instructions
   - Performance and security monitoring setup for supply chain operations

2. **Technical Implementation Documentation (1,500-2,000 words)**

   Architecture Documentation:

   - Supply chain system architecture with component interaction analysis focusing on timer services and logistics security
   - Implementation specifications with supply chain transaction management and interceptor strategies
   - Configuration documentation with logistics security hardening and performance tuning parameters
   - Deployment architecture with split directory structure and packaging considerations for supply chain modules

   Component Analysis:

   - Timer service implementations with supply chain scheduling strategies and persistence analysis
   - Interceptor framework design with logistics performance impact assessment
   - Transaction management design with supply chain data isolation and recovery strategies
   - Logistics security architecture with authentication, authorization, and vendor session management
   - Exception handling framework with supply chain recovery and optimization mechanisms

Implementation Analysis with Critical Evaluation:

- Critical analysis of timer service design decisions with global supply chain clustering considerations
- Interceptor implementation strategy with logistics performance impact assessment
- Transaction management approach with supply chain attribute selection justification
- Logistics security architecture evaluation with vendor data threat mitigation strategies
- EJB best practices implementation with supply chain performance optimization analysis
- Exception handling strategy with logistics resilience and recovery evaluation

3. **Critical Analysis and Test Report (1,000-1,500 words)**

Testing Strategy and Performance Validation:

- Unit testing approach with timer service and supply chain transaction testing
- Integration testing with logistics security and interceptor validation
- Performance benchmarking with timer service and supply chain transaction optimization results
- Logistics security testing with penetration testing and vendor data vulnerability assessment
- Exception handling testing with supply chain failure scenario validation
- Critical evaluation of testing results with global logistics improvement recommendations

Critical Reflection and Optimization:

- Analysis of timer service alternatives with justification for chosen supply chain approaches
- Performance bottleneck identification in interceptor and logistics transaction processing
- Supply chain security analysis with threat Modeling and vendor data mitigation strategies
- Critical assessment of exception handling effectiveness with logistics optimization strategies
- Deployment strategy evaluation with packaging and global supply chain performance considerations

-END OF ASSIGNMENT-

Evaluation Criteria

- **Critical Analysis and Technical Understanding (40 marks)**
  - **Timer Services Implementation and Integration Analysis (6 marks)**
    Demonstrates comprehensive understanding of EJB timer services with effective business process integration and critical evaluation of timer reliability and performance.
  - **Interceptor Architecture and Business Logic Management (6 marks)**
    Showa mastery of interceptor concepts with multiple interceptor integration and critical analysis of cross-cutting concern implementation.
  - **Transaction Management and Demarcation Strategy (6 marks)**
    Demonstrates clear understanding of transaction types with appropriate scenario-based implementation and critical evaluation of transaction strategies.
  - **Security Architecture and Programmatic Authorization (6 marks)**
    Shows comprehensive security framework implementation with effective programmatic authorization and critical analysis of session privacy measures.
  - **EJB Best Practices and Performance Optimization (8 marks)**
    Demonstrates mastery of EJB best practices with effective exception handling optimization and critical evaluation of performance enhancement strategies.
  - **Professional Deployment and Packaging Architecture (8 marks)**
    Shows professional-level deployment organization with appropriate packaging strategies and critical analysis of deployment architecture effectiveness.

- **Working Implementation Quality (35 marks)**
  - **Functional Advanced EJB Features Implementation (10 marks)**
    Complete implementation of timer services, interceptors, transaction management, and security features with integrated performance monitoring.
  - **Correct Implementation of Required EJB Components (8 marks)**
    Proper implementation and integration of advanced EJB features as per assignment requirements with professional code quality.
  - **Performance Optimization and Exception Handling (9 marks)**
    Effective exception handling implementation with demonstrated performance optimization and EJB-specific exception management.
  - **Professional Deployment and Package Structure (8 marks)**
    Professional deployment architecture with proper packaging, split directory structure, and environment configuration management.

- **Testing and Validation (25 marks)**
  - **Comprehensive Testing Strategy for Advanced Features (8 marks)**
    Well-defined testing plan including timer service testing, interceptor validation, transaction testing, security validation, and performance testing.
  - **Test Execution and Advanced Feature Validation (6 marks)**
    Evidence of comprehensive testing with meaningful analysis of advanced EJB feature effectiveness and performance.
  - **Performance and Security Optimization Evaluation (6 marks)**
    Critical evaluation of performance optimization through exception handling and security implementation effectiveness.
  - **Professional Testing Documentation and Enterprise Recommendations (5 marks)**
    Professional testing documentation with enterprise-level recommendations for continuous improvement and optimization.

Assignment Submission Guidelines for Students

- **Format:**
  - Assignment Documents Rules:
    - Paper Size: A4
    - Line Spacing:1.5
    - Header and Footer:1 inch
    - Basic Font Size:11pt
    - Heading:14pt
    - Sub-headings:12pt, Bold
    - Body:11pt, Justified Aligned
    - Font Style: Times New Roman/Calibri
  - Provide the following information in their assignment coversheet:
    - Student name
    - NIC No
    - Subject name
    - Subject code
    - Branch
  - Working project as compressed archive with complete implementation
  - Combined documentation as single PDF (2,500-3,500 words total)
  - Performance monitoring results and optimization evidence
- **Code Requirements:** Must compile, deploy, and demonstrate performance optimization
- **Individual Work:** Original implementation and critical analysis required
- **Resources:** Official Java EE documentation and academic sources with proper citation
- **Referencing:** Must be cited using the Harvard referencing system, with a reference list included at the end

Submission Checklist:

- ☐ Complete source code with timer services, interceptors, and security implementation
- ☐ Deployable supply chain application with split directory structure
- ☐ Database setup with transaction coordination and logistics audit trail
- ☐ Comprehensive test suite with security and performance validation
- ☐ Critical analysis documentation with timer service and security evaluation
- ☐ Performance monitoring and security testing results

Academic Integrity and Assessment Notes

Critical Thinking Emphasis: This assignment prioritizes critical analysis and evaluation of timer services, security architecture, and transaction management over comprehensive implementation. Deep understanding of EJB concepts with effective optimization strategies may receive higher marks than extensive implementations without critical analysis.

Security and Performance Focus: The systematic analysis of security measures, timer service optimization, and exception handling demonstrates understanding of enterprise supply chain challenges. Quality of critical thinking and problem-solving approach is essential for success.

Real-World Application: Your solution should demonstrate practical understanding of global supply chain system requirements with critical evaluation of design decisions, security trade-offs, and optimization strategies suitable for production logistics environments.